



Stable Matchings & Hall's Theorem

Lecture 7 | Algorithms Design & Analysis

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Recall: Gale-Shapley Algorithm



Repeat until all men are matched

1. Each woman proposes to most preferred man
2. Each man tentatively chooses the most preferred woman from whom he got proposals and rejects the others.
3. If a man who was tentatively matched previously, gets a better offer according to his preference list, he switches.
4. Woman who gets reject, removes that man from their preference list.

Optimal for Woman



Theorem: Algorithm results in every woman getting her best choice among all possible stable matchings.

Theorem: Algorithm results in every man getting his worst choice among all possible stable matchings.

Proofs: Page 10 in Klienberg-Tardos:

<https://theswissbay.ch/pdf/Gentoomen%20Library/Algorithms/Algorithm%20Design%20-%20John%20Kleinberg%20-%20%C3%89va%20Tardos.pdf#page=13.40>

Read More: Popular Matchings



Watch: <https://www.youtube.com/watch?v=nbaRTJtYwuk>

Hall's Theorem



Theorem 2. *For a bipartite graph G on the parts X and Y , the following conditions are equivalent.*

- (a) *There is a perfect matching of X into Y .*
- (b) *For each $T \subseteq X$, the inequality $|T| \leq |N_G(T)|$ holds.*

Proof:

a.) $\Rightarrow$ b.) : Easy

b.) $\Rightarrow$ a.) : 2 Cases

Case 1: b is satisfied with strict inequality.

Easy

Case 2: There is some T for which b is satisfied only with equality

See:

<https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/30.%20Matchings%20and%20Hall's%20Theorem/Matching%20and%20Hall's%20Theorem.pdf>

Read



For Stable Matchings
Kleinberg Tardos, Chapter 1

For Hall's Theorem Proof:

<https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/30.%20Matchings%20and%20Hall's%20Theorem/Matching%20and%20Hall's%20Theorem.pdf>

Solve



Exercises 1, 2 in

<https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/30.%20Matchings%20and%20Hall's%20Theorem/Matching%20and%20Hall's%20Theorem.pdf>